

CHERNUSHKA, Russian Federation

WMO: 284280

Lat: 56.500N

Lon: 56.133E

Elev: 487

StdP: 14.44

Time Zone: 5.00 (E05)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.7	-19.7	-31.0	1.0	-25.7	-24.6	1.4	-19.5	22.7	20.2	20.4	20.5	2.9	180	0.614

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	86.3	68.7	82.9	67.1	79.7	65.6	70.8	82.7	69.1	79.9	67.3	76.9	6.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.7	100.0	76.2	65.1	94.5	74.5	63.5	89.4	72.6	34.9	82.9	33.5	80.6	32.0	76.7	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.8	17.5	15.5	-34.9	90.2	6.6	4.0	-39.7	93.1	-43.6	95.4	-47.3	97.7	-52.1	100.6		
			-35.1	73.5	6.6	1.8	-39.9	74.8	-43.7	75.8	-47.5	76.8	-52.3	78.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.1	7.1	9.0	22.7	38.4	53.2	61.3	65.3	61.5	51.3	37.9	24.8	10.6		
	DBStd	23.11	14.18	13.42	9.11	9.44	8.45	7.69	6.89	7.43	7.56	7.95	10.61	14.36		
	HDD50	6198	1331	1149	847	364	65	9	0	4	71	381	757	1220		
	HDD65	10390	1796	1569	1312	797	372	158	83	157	415	839	1207	1685		
	CDD50	1486	0	0	0	18	165	348	476	362	110	7	0	0		
	CDD65	201	0	0	0	0	8	47	93	50	3	0	0	0		
	CDH74	2363	0	0	0	7	174	543	1037	555	47	0	0	0		
	CDH80	662	0	0	0	0	23	141	318	173	7	0	0	0		
	WSAvg	6.7	7.2	7.3	7.9	7.0	7.0	6.0	5.0	5.2	5.9	7.3	7.6	7.2		
	PrecAvg	23.1	1.5	1.1	1.2	1.1	1.7	2.8	2.8	2.5	2.3	2.5	2.0	1.6		
	PrecMax	30.2	3.1	2.2	2.2	2.6	3.0	5.9	6.0	5.2	6.0	4.4	3.3	2.7		
Precipitation	PrecMin	15.4	0.3	0.4	0.1	0.0	0.7	0.7	0.6	0.6	0.7	0.6	0.4	0.5		
	PrecStd	3.2	0.6	0.5	0.6	0.7	0.7	1.3	1.5	1.1	1.2	1.0	0.9	0.6		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.9	35.0	45.4	73.2	82.9	89.4	90.9	90.6	80.2	64.8	48.9	34.9		
	0.4%	MCWB	32.8	32.4	37.8	56.1	62.2	69.4	69.8	68.8	60.8	52.0	45.9	33.5		
	2%	DB	31.7	33.1	40.5	65.6	79.1	83.7	86.7	84.8	72.9	58.5	43.8	33.6		
	2%	MCWB	30.7	31.7	35.3	52.4	60.7	67.0	69.4	68.2	60.0	48.8	42.0	32.7		
	5%	DB	29.6	31.2	37.3	58.9	74.9	80.1	83.5	80.0	68.0	53.6	39.5	32.1		
	5%	MCWB	28.7	29.9	33.3	47.9	58.6	66.2	68.6	66.2	58.0	47.2	38.0	31.0		
	10%	DB	26.4	27.7	34.8	53.1	70.5	76.4	80.1	75.6	63.8	49.7	36.0	29.4		
	10%	MCWB	25.3	26.1	32.0	45.0	56.8	64.2	67.3	64.4	55.7	46.2	34.5	28.2		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.3	33.5	39.2	57.1	65.9	71.7	73.1	71.3	64.2	53.6	46.4	34.2		
	0.4%	MCDB	33.8	34.2	44.0	71.4	78.3	84.7	84.5	85.0	72.9	61.7	48.7	34.4		
	2%	WB	30.9	31.9	36.2	53.3	62.6	69.4	71.5	69.3	61.1	50.3	42.0	32.9		
	2%	MCDB	31.4	33.0	39.4	63.8	74.8	80.2	83.2	81.4	69.7	55.5	43.4	33.5		
	5%	WB	28.7	29.8	34.1	49.3	60.1	67.4	70.0	67.2	58.9	48.2	38.1	31.0		
	5%	MCDB	29.4	31.2	36.3	57.6	71.9	77.4	81.0	77.6	66.9	52.0	39.3	32.0		
	10%	WB	25.4	26.3	32.4	45.5	57.8	65.4	68.2	65.3	56.8	46.2	34.9	28.3		
	10%	MCDB	26.5	27.6	34.4	52.4	69.1	74.4	77.9	74.3	62.9	49.8	35.8	29.5		
Mean Daily Temperature Range		MDBR	13.1	15.7	16.0	17.0	21.3	20.1	20.2	19.1	17.0	10.9	8.6	11.3		
	5% DB	MCDBR	9.5	10.3	16.4	25.8	29.3	26.3	26.1	27.0	24.0	20.9	8.7	7.4		
		MCWBR	9.6	9.5	12.2	14.3	13.6	12.2	11.1	11.6	12.3	12.8	7.6	7.5		
	5% WB	MCDBR	9.3	9.7	12.2	24.0	25.7	23.3	23.2	24.5	21.0	18.5	8.5	8.1		
		MCWBR	9.5	9.1	9.5	13.8	13.4	12.7	11.0	11.7	11.4	12.7	7.9	8.2		
Clear-Sky Solar Irradiance	taub		0.296	0.320	0.375	0.409	0.398	0.393	0.414	0.435	0.381	0.355	0.315	0.293		
	taud		2.007	2.012	1.923	2.053	2.265	2.336	2.286	2.203	2.323	2.257	2.113	1.978		
	Ebn at Noon		172	222	238	253	266	269	260	244	242	212	166	136		
	Edh at Noon		21	31	44	45	39	37	38	39	30	24	19	16		
All-Sky Solar Radiation	RadAvg		210	510	944	1341	1661	1760	1768	1350	829	362	169	120		
	RadStd		29	71	95	162	178	169	191	129	78	72	35	27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page